

EXP NO: 8

SHARED PREFERENCES

AIM:

To implement the shared preference in Android Application.

ALGORITHM:

STEP 1: The first thing we need to do is to create one shared preferences file per app.

STEP 2: Name it with the package name of your app- unique and easy to associate with the app.

STEP 3: When you want to get the values, call the `getSharedPreferences()` method.

STEP4: To write to a SharedPreference file we can create a `SharedPreferences.Editor` by calling `edit()` on your shared preference.

STEP 5: To retrieve values from a shared preferences file, call methods such as `getInt()` and `getString()`

CODE: XML

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"
    tools:ignore="HardcodedText">
    <TextView
        android:id="@+id/textview"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="32dp"
        android:text="Shared Preferences Demo"
        android:textColor="@android:color/black"
        android:textSize="24sp" />
    <!--EditText to take the data from the user and save the data in SharedPreferences-->
    <EditText
        android:id="@+id/edit1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_below="@+id/textview"
        android:layout_marginStart="16dp"
        android:layout_marginTop="8dp"
        android:layout_marginEnd="16dp"
        android:hint="Enter your Name"
        android:padding="10dp" />
```

<!--EditText to take the data from the user and save the data in
SharedPreferences-->

```
<EditText
    android:id="@+id/edit2"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@+id/edit1"
    android:layout_marginStart="16dp"
    android:layout_marginTop="8dp"
    android:layout_marginEnd="16dp"
    android:hint="Enter your Age"
    android:inputType="number"
    android:padding="10dp" />
```

</RelativeLayout>

CODE: MainActivity.java

```
import androidx.appcompat.app.AppCompatActivity;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.widget.EditText;
```

```
public class MainActivity extends AppCompatActivity {
    private EditText name, age;
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        name = findViewById(R.id.edit1);
        age = findViewById(R.id.edit2);
    }
```

// Fetch the stored data in onResume() Because this is what will be called when
the app opens again

```
    @Override
    protected void onResume() {
        super.onResume();
        // Fetching the stored data from the SharedPreferences
```

```

        SharedPreferences sh = getSharedPreferences("MySharedPref",
MODE_PRIVATE);
        String s1 = sh.getString("name", "");
        int a = sh.getInt("age", 0);

        // Setting the fetched data in the EditTexts
        name.setText(s1);
        age.setText(String.valueOf(a));
    }

    // Store the data in the SharedPreferences in the onPause() method
    // When the user closes the application onPause() will be called and data will be
stored
    @Override
    protected void onPause() {
        super.onPause();
        // Creating a shared pref object with a file name "MySharedPref" in private
mode
        SharedPreferences sharedPreferences =
getSharedPreferences("MySharedPref", MODE_PRIVATE);
        SharedPreferences.Editor myEdit = sharedPreferences.edit();

        // write all the data entered by the user in SharedPreferences and apply
myEdit.putString("name", name.getText().toString());
myEdit.putInt("age", Integer.parseInt(age.getText().toString()));
myEdit.apply();
    }
}

```

OUTPUT:

10:15

Shared Preferences Demo

Enter your Name

Enter your Age

1 2 3 -

4 5 6 ,

7 8 9 ✕

. 0 ✓

This screenshot shows the initial state of the application. The title bar is purple and displays the time 10:15. The app title 'Shared Preferences Demo' is centered at the top. Below it are two text input fields with placeholder text 'Enter your Name' and 'Enter your Age'. At the bottom, there is a light gray numeric keypad with digits 1-9, 0, a decimal point, a minus sign, a comma, a delete icon (✕), and a green checkmark icon (✓). The Android navigation bar is visible at the very bottom.

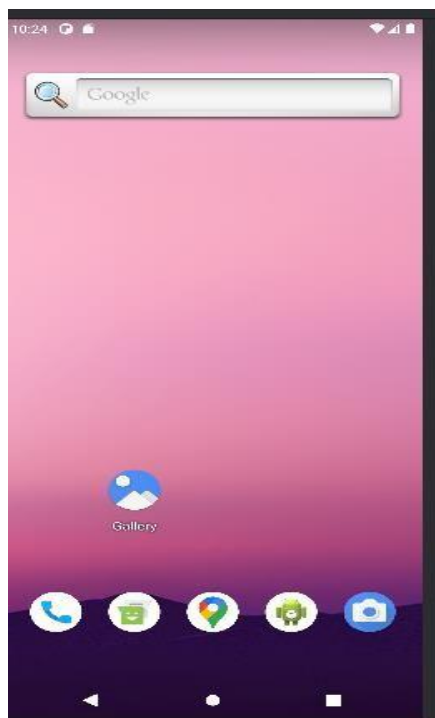
10:14

Shared Preferences Demo

Sujitha

21

This screenshot shows the application after data entry. The title bar now displays the time 10:14. The input fields are populated with the text 'Sujitha' and the number '21'. The numeric keypad is no longer visible, indicating it has been dismissed. The Android navigation bar is visible at the bottom.



RESULT:

Thus, The Android Application has been developed and run successfully.